

INPPL1 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes INPPL1 gene fusions from the msk_impact_50k_2026 study, covering 14 fusion events. 4/14 fusions (28.6%) were in-frame. 7/14 fusions (50.0%) retained the PF03372 domain. Domain retention was tested with Fisher's exact test ($p=0.048951$, statistically significant at $\alpha=0.05$). No literature benchmark is configured for INPPL1.

1 Results summary

1.1 Fusion partner frequency

Fusion-partner frequency was tabulated across 14 analyzed fusion events, identifying 11 distinct fusion partner genes. The most frequent partner was FCHSD2, observed in 4 events.

1.2 Domain retention

PF03372 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 4/4 (100.0%) of in-frame fusions retained the domain, $p=0.048951$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.108911.

1.3 Domain disruption

Domain-disruption analysis was skipped: INPPL1 has no `disruption_required_domains` configured; domain-disruption analysis was skipped.

1.4 Cutpoint detection

Cutpoint detection scanned 13 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 560 aa (permutation-corrected $p=0.033966$, statistically significant at $\alpha=0.05$). This is 131 aa from the nearest configured domain boundary at 429 aa.

1.5 Corroborating confidence statistics

Corroborating confidence statistics were not computed for this run: INPPL1 has no `group_field` configured; confidence-stats analysis was skipped.

1.6 Composite evidence score

Composite evidence score reported: `domain_disruption` sub-score excluded: the supplied result had no applicable p-value (e.g. `disruption_required_domains` not configured for this gene).

1.7 Exon retention

Exon retention reported: No target exon was supplied in `params` or `GeneConfig`.

1.8 Joint partner

Joint partner reported: INPPL1 has no `gene_pair` configured; joint-partner analysis was skipped.

2 Tables

2.1 composite_score composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty	Components_applicable	Composite_score	Rank
ANK2	1	0.07142857142857142	0.09629286886244175		1.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2792882694916884	1
SHANK2	1	0.07142857142857142	0.09629286886244175		0.956		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.26985969806311694	2
FCHSD2	4	0.2857142857142857	0.09629286886244175		0.51		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2661250041855659	3
PTCH1	1	0.07142857142857142	0.09629286886244175		0.608		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.1952882694916884	4
GAB2	1	0.07142857142857142	0.09629286886244175		0.31999999999999995		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.13357398377740265	5
NLRP6	1	0.07142857142857142	0.09629286886244175		0.29200000000000004		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.12757398377740267	6
CLPB	1	0.07142857142857142	0.09629286886244175		0.246		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.11771684092025979	7
FOLR2	1	0.07142857142857142	0.09629286886244175		0.14600000000000002		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.09628826949168838	8
FAT3	1	0.07142857142857142	0.09629286886244175		0.10799999999999998		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.08814541234883122	9
SLC67A1	1	0.07142857142857142	0.09629286886244175				['domain_retention', 'recurrence']	0.08273052480760337	10
LRTOMT	1	0.07142857142857142	0.09629286886244175		0.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.06500255520597409	11

2.2 cutpoint_detection cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
114	0	7	1	5	0.0	0.46153846153846156	0.3357921019231931
133	0	7	2	4	0.0	0.1923076923076923	0.7160033436347992
183	1	6	2	4	0.3333333333333333	0.5594405594405595	0.2522460504731182
206	1	6	3	3	0.16666666666666666	0.26573426573426573	0.5755524408482516
220	1	6	4	2	0.08333333333333333	0.10256410256410256	0.9890046156985368
364	2	5	4	2	0.2	0.2861305361305362	0.5434357913897181
538	2	5	5	1	0.08	0.10256410256410256	0.9890046156985368
560	2	5	6	0	0.0	0.020979020979020976	1.6782147827453995
805	6	1	6	0	0.0	1.0	-0.0

2.3 domain_retention frame_domain_contingency_table

4	3
0	6

2.4 domain_retention domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained
auto_key_domain	Endonuclease/Exonuclease/phosphatase family	13	7	2	4	0.17988505747126438

2.5 frequency Partner_gene_counts

Partner_gene	Event_count
ANK2	1
CLPB	1
FAT3	1
FCHSD2	4
FOLR2	1
GAB2	1
LRTOMT	1
NLRP6	1
PTCH1	1
SHANK2	1
SLC67A1	1

2.6 Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_5prime	breakpoint_3prime	breakpoint_5prime	breakpoint_3prime	is_retained	is_lost	domain_retention	domain_retention	domain_retention	disrupted_domain	sequence_annotation	effect_on_fusion
EVT-P-0007343-T02-IM6-4	P-0007343-T02-IM6		INPPL1-ANK2	ANK2	unknown	five_prime	71943348	14	560		False	disrupted	0.45517241379310347	True	False	Endonuclease family	ENPPL1 (NM_001567) - ANK2 (NM_001148) rearrangement: t(4;11)(q25;q26;q13.4)(chr4:g.113898703::chr11:g.71943348) Protein Fusion: mid-exon {INPPL1:ANK2}	NA
EVT-P-0031523-T02-IM6-5	P-0031523-T02-IM6		INPPL1-CLPB	CLPB	unknown	three_prime	71940164	5	183		False	retained	1.0	False	False	Endonuclease/Exonuclease/phosphatase family	INPPL1 (NM_001567) rearrangement: c.549:INPPL1_chr11:g.72024501del Antisense Fusion: mid-exon {INPPL1:ANK2}	NA
EVT-P-0042110-T02-IM7-6	P-0042110-T02-IM7		INPPL1-FAT3	FAT3	unknown	five_prime	71939487	3	114		False	lost	0.0	False	False	Endonuclease/Exonuclease/phosphatase family	INPPL1 (NM_001567) - FAT3 (NM_001008781) rearrangement: c.342:INPPL1_c.3669+26772:FAT3del Protein Fusion: mid-exon {INPPL1:FAT3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_coding	breakpoint_4	main_retained	donor_frame	main_retained	donor_frame	disrupted_domain	description_annotation	effect_on_fusion
EVT-P-0004226-T08-IM7-7	P-0004226-T08-IM7		INPPL1-FCHSD2	FCHSD2	in-frame	five_prime	71945806	21	805	True	retained	1.0	False			Endonuclease/Exonuclease/phosphatase family	INPPL1 (NM_001567) - FCHSD2 (NM_014824) rearrangement: c.2415+147:INPPL1_c.705+16442:FCHSD2dup Protein Fusion: in frame {INPPL1:FCHSD2}	705+16442:FCHSD2dup
EVT-P-0004226-T04-IM6-8	P-0004226-T04-IM6		INPPL1-FCHSD2	FCHSD2	in-frame	five_prime	71945806	21	805	True	retained	1.0	False			Endonuclease/Exonuclease/phosphatase family	INPPL1 (NM_001567) - FCHSD2 (NM_014824) rearrangement: c.2415+147:INPPL1_c.705+16442:FCHSD2dup Protein Fusion: in frame {INPPL1:FCHSD2}	705+16442:FCHSD2dup
EVT-P-0004226-T03-IM6-9	P-0004226-T03-IM6		INPPL1-FCHSD2	FCHSD2	in-frame	five_prime	71945742	21	805	True	retained	1.0	False			Endonuclease/Exonuclease/phosphatase family	INPPL1 (NM_001567) - FCHSD2 (NM_014824) rearrangement: c.2415+83:INPPL1_c.705+16440:FCHSD2dup Protein Fusion: in frame {INPPL1:FCHSD2}	705+16440:FCHSD2dup
EVT-P-0004226-T07-IM7-10	P-0004226-T07-IM7		INPPL1-FCHSD2	FCHSD2	in-frame	five_prime	71945792	21	805	True	retained	1.0	False			Endonuclease/Exonuclease/phosphatase family	INPPL1 (NM_001567) - FCHSD2 (NM_014824) rearrangement: c.2415+133:INPPL1_c.705+16441:FCHSD2dup Protein Fusion: in frame {INPPL1:FCHSD2}	705+16441:FCHSD2dup
EVT-P-0045578-T01-IM6-11	P-0045578-T01-IM6		INPPL1-FOLR2	FOLR2	out-of-frame	five_prime	71939641	3	133	True	lost	0.0	False			Endonuclease/Exonuclease/phosphatase family	INPPL1 (NM_001567) - FOLR2 (NM_001113536) rearrangement: c.397+99:INPPL1_c.151-191:FOLR2dup Protein Fusion: out of frame {INPPL1:FOLR2}	151-191:FOLR2dup
EVT-P-0009791-T02-IM6-12	P-0009791-T02-IM6		INPPL1-GAB2	GAB2	out-of-frame	five_prime	71940493	6	220	True	lost	0.0	False			Endonuclease/Exonuclease/phosphatase family	INPPL1 (NM_001567) - GAB2 (NM_080491) rearrangement: c.660-16:INPPL1_c.75+18325:GAB2inv Protein Fusion: out of frame {INPPL1:GAB2}	75+18325:GAB2inv
EVT-P-0048151-T01-IM6-31	P-0048151-T01-IM6		INPPL1-LRTOMT	LRTOMT	unknown	five_prime	71948466	26	1060	False	retained	1.0	False			Endonuclease/Exonuclease/phosphatase family	INPPL1 (NM_001567) - LRTOMT (NM_145309) rearrangement: c.3178:INPPL1_c.83-1241:LRTOMTdup Protein Fusion: mid-exon {INPPL1:LRTOMT}	83-1241:LRTOMTdup

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint	splice_point	exon breakpoint	protein_pos	domain_pos	main_retained	donor_intron	invariant_deleted	lost_domain	disrupted_domains	sequence_annotation	transcript_effect_on_frame
EVT-P-0005693-T03-IM6-34	P-0005693-T03-IM6		INPPL1-PTCH1	PTCH1	out-of-frame	three_prime	71941318	9	364	True	retained	1.0	False	Endonuclease family			PTCH1 (NM_000264) - INPPL1 (NM_001567) rearrangement: t(9;11)(q22.32;q13.4)(chr9:g.98211830::c.11831T>C chr11:g.113111111T>A) Protein Fusion: out of frame {PTCH1:INPPL1}	
EVT-P-0024858-T01-IM6-36	P-0024858-T01-IM6		NLRP6-INPPL1	NLRP6	unknown	five_prime	71940232	5	206	False	lost	0.0	False	Endonuclease family			NLRP6 (NM_001567) - INPPL1 (NM_138329) rearrangement: c.617:INPPL1_c.2138:NLRP6dup Protein Fusion: mid-exon {INPPL1:NLRP6}	
EVT-P-0031882-T01-IM6-37	P-0031882-T01-IM6		SHANK2-INPPL1	SHANK2	unknown	three_prime	71942656	13	538	False	disrupted	0.62413793	0.04628	Endonuclease family			SHANK2 (NM_001567) - INPPL1 (NM_001567) rearrangement: c.1612:INPPL1_chr11:g.70501515dup An-tisense Fusion	

3 Figures

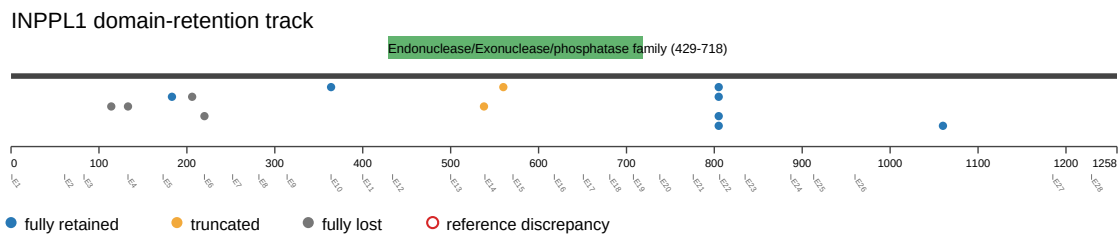


Figure 1: domain retention outliers

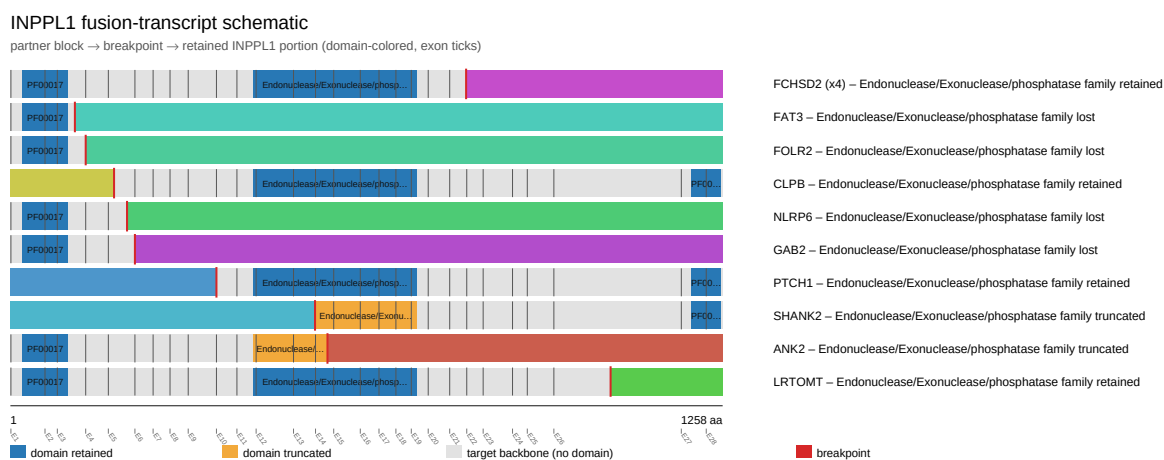


Figure 2: fusion schematic

Reference vs msk_impact_50k_2026



Figure 3: reference comparison